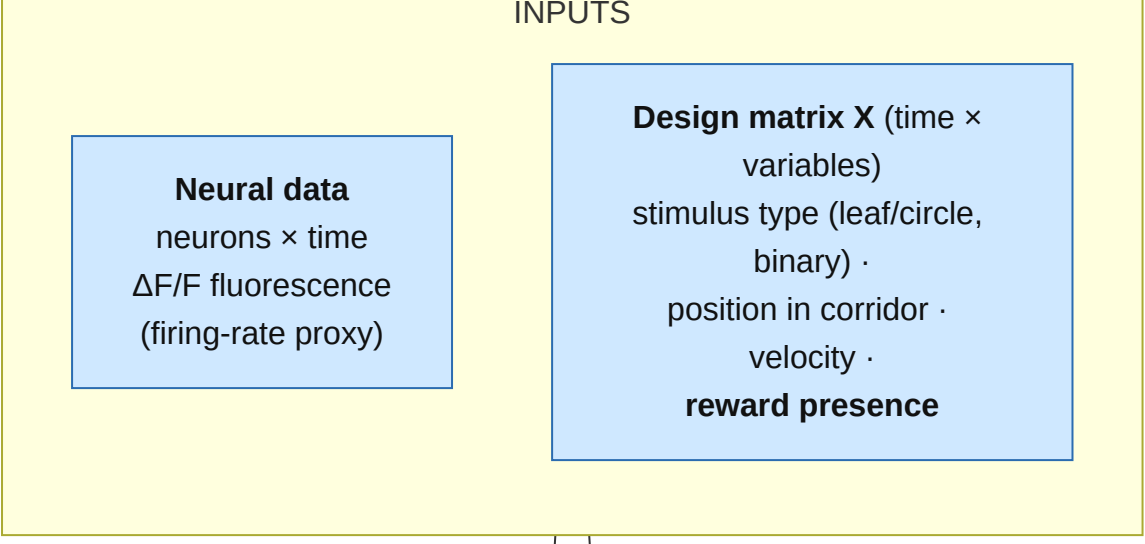


GOAL
Identify **reward-encoding neurons** and track how their proportion changes across training days, by region, cohort, and stimulus type.



PREPROCESS
bin ΔF into time bins, take mean per bin;
align neural data and X on the same time axis

SPLIT BY TRIAL
80% train / 20% test
(split by trial, not by bin — bins within a trial are correlated)

FIT FULL MODEL (per neuron)
 $y = X\theta$; estimate θ by least squares (Gaussian MLE) on train

TEST ERROR (full)
predict $\hat{y}$ on held-out 20%;
 $\text{error_full} = \text{MSE}(y, \hat{y})$

OMISSION / ABLATION
drop the reward column from X, refit, predict;
 $\text{error_reduced} = \text{MSE}$ on held-out 20%

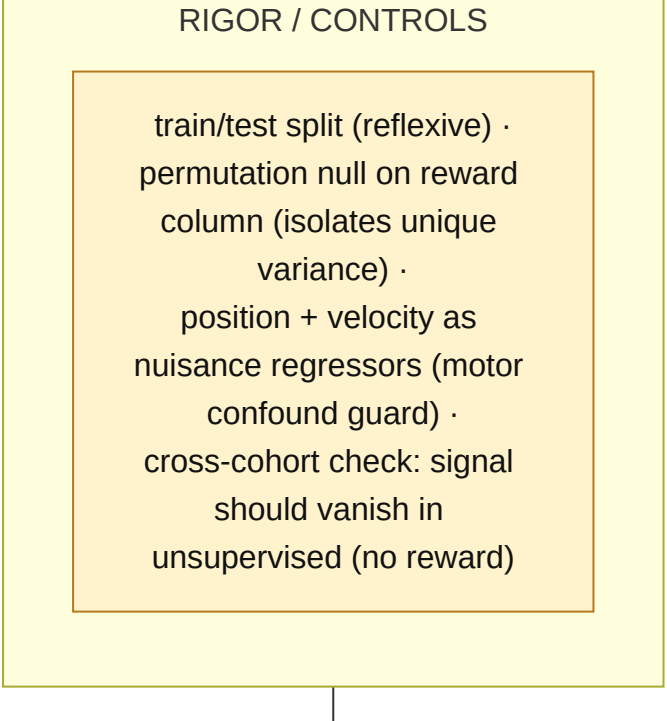
EFFECT SIZE (per neuron)
 $d = \text{error_reduced} - \text{error_full}$
 $d > 0 \Rightarrow$ reward improved the fit

PERMUTATION TEST
shuffle the **reward column** across trials, refit, recompute d;
repeat ×1000 → null distribution of d

p-VALUE
 $p = \text{fraction of shuffled } d \geq \text{true } d$
(small steady d can beat a large noisy one)

CLASSIFY
neuron is 'reward-encoding' if $p < \text{threshold}$

AGGREGATE
proportion of reward-encoding neurons:
across **training days** · by **brain region** ·
supervised vs unsupervised · **leaf vs circle**



INTERPRETATION
Significant d $\Rightarrow$ trustworthy reward attribution.
Null d $\Rightarrow$ **ambiguous** — reward may matter but be too collinear with stimulus to separate.